Coordination Without a Coordinator: A Scientific Programme for Hormone-Inspired Residual-Demand Signalling

Tilanthi
BIODISC — Biology Discovery and Intelligence System

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ABSTRACT

How much must a node know, and what must a shared medium provide, for n minimally-informed autonomous agents to produce a specified global behaviour — reliably, terminably, and with coordination cost carried by an explicit five-component vector (agent and medium communication, agent and medium state, latency; the agent-side component $O(n\,k)$ with $k \ll n$, conditional on a medium providing broadcast, locality and decay) — with no central coordinator? Mammalian wound healing is an existence proof over a substantial operating envelope: a wound is repaired by cells that perceive only local chemistry plus built-in capability, organised end-to-end by broadcast signals that receivers interpret, with production of new responders ramped by the unconsumed remainder of a constitutive signal. We distil this reference system into five coordination primitives — receiver-defined meaning, concentration-as-instruction, residual-signal-as-demand, co-stimulus guards, antagonist termination — plus one separated architectural assumption (cheap responders, elastic factories), and find that the sixteen families we survey jointly fail to cover them, the residual-signal demand loop above all. We derive the central law's equilibria and its delay-stability threshold, demonstrate its fibrotic failure mode in simulation, and lay out a five-phase programme — controller analysis, factorial ablation sandbox, benchmarks and inverse design, substrate engineering, and failure theory mapping the operating boundary — with falsifiable predictions and stated criteria per phase (a termination criterion for the core mechanism, claim-reducing criteria elsewhere), testing three separable hypotheses: mechanistic, architectural, and comparative. The medium is not free, and whether the coordinator has merely been hidden inside it is a criterion the programme itself tests, not an assumption.

Key words: distributed systems — self-organization — bio-inspired computing — amorphous computing — wound healing — autonomous systems

1 INTRODUCTION

This paper is organised around one basic question:

How much must a node know, and what must a shared medium provide, for n minimally-informed autonomous agents to produce a specified global behaviour — reliably, terminably, and at communication cost $O(n\,k)$ with $k \ll n$ — with no central coordinator?

The premise hiding in the question is economic. If every agent must coordinate with every other, coordination state grows as $O(n^2)$ and compounds with growth. If each agent holds only local state and reacts to signals by built-in capability, a disturbance involving k agents costs roughly $O(n\,k)$ in signalling, and adding an agent adds capacity rather than coordination load. The premise is not free: it requires a medium providing broadcast with locality and decay, and it restricts which behaviours are reachable; the programme makes those requirements precise. The paper's cost statement is accordingly a five-component vector — agent and medium communication, agent and medium state, latency — of which $O(n\,k)$ is only the agent-communication component; the medium's share is a real cost paid in Phase IV, not an idealisation. One distinction organises everything that follows: what is decentralised is *decision-making and knowledge*, not necessarily physical plant — factories may be centralised and media shared; the argument is against centralised *deciding*, not against shared infrastructure (Section 5).

Biology answers the question once, in tissue repair. Exposed blood triggers thrombin, which polymerises fibrin and traps red cells into a mat that stops the bleeding — local chemistry, no coordinator. Damaged cells release stress signals that recruit neutrophils and macrophages to clear contamination (Medzhitov 2008). If local responders are insufficient, the unconsumed remainder of the hormone thrombopoietin accumulates in plasma and ramps platelet production in the marrow (Kaushansky 2005); G-CSF runs a related, further-layered programme for neutrophils under severe demand (Manz & Boettcher 2014). Every cell does what it is programmed to do; signals arrive with the blood; the wound closes. The same mechanism operates from pinprick to major trauma — though overwhelming injury does break it, a

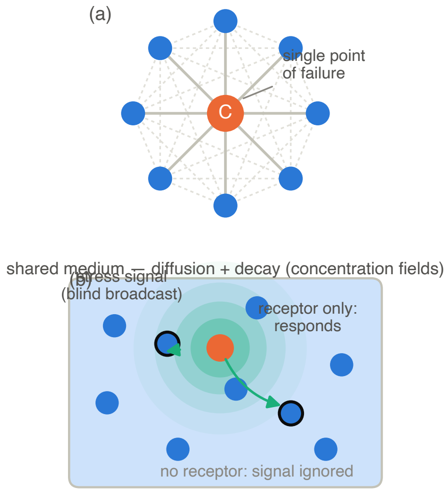

Figure 1. Two coordination architectures. (a) Central orchestration: the hub holds $O(n)$ state and pairwise traffic grows as $O(n^2)$; the hub is a single point of failure. (b) A hormonal medium: a stressed agent emits a signal that diffuses and decays; only receptor-equipped agents respond. Every agent holds $O(1)$ state; a disturbance reaching k receivers costs $O(n\,k)$ messages. Meaning lives in the receiver.

boundary the programme must map rather than assume away (Gurtner et al. 2008).

We propose to treat this as more than an analogy: read wound healing as an *engineering specification* that evolution has stress-tested — robust across a substantial envelope, though neither optimal nor pathology-free — extract its coordination principles (Section 3), and subject them to the scrutiny any competing architecture would face (Section 4); Fig. 1 contrasts the two architectures.

The contributions are: (i) a precise statement of the coordinator-free coordination question, with the five-component cost vector (agent and medium communication, agent and medium state, latency) as the primary complexity claim, of which $O(n\,k)$ is the agent-side component; (ii) a five-primitive distillation of wound healing plus one separated architectural assumption, with its pathologies (cytokine storm, thrombosis, autoimmunity, fibrosis) read as engineering safeguards; (iii) a gap analysis showing the sixteen families surveyed jointly fail to cover the primitives — the residual-signal demand loop absent from all of them; and (iv) a five-phase programme with falsifiable predictions and a stated criterion per phase — terminal for the core mechanism, claim-reducing elsewhere — organised around three hypotheses that can fail independently (Section 4). This is a research-programme paper: it reports an equilibrium and stability analysis of the central control law and one illustrative simulation (Fig. 3), and a falsifiable plan — not completed experiments; Phase II onward is where the predictions are tested. The predictions, effect sizes and kill criteria are written to be registrable: a Perspective or Stage-1 registered-report track is the intended submission form.

2 PRIOR APPROACHES

2.1 Central orchestration, autonomic and control-theoretic computing

The dominant engineering answer is a coordinator: cluster managers such as Borg and Kubernetes concentrate placement, scaling and failure handling at a (replicated) control plane (Burns et al. 2016), autonomic computing envisioned managed systems reporting to an external control loop (Kephart & Chess 2003), and control-theoretic computing designs classical feedback controllers against measured metrics (Hellerstein et al. 2004). These work at impressive scale, but the coordination is bought: global state, global visibility, superlinear traffic. The modern ecosystem — service-mesh sidecars, event-driven autoscaling, chaos engineering — refines this pipeline without changing its epistemics: the error signal is still measured and reported, never physical. The impossibility result of Fischer et al. (1985) — consensus with one faulty process is unsolvable asynchronously — directed a large literature towards tolerating coordinators rather than eliminating them. Our question is orthogonal: not how to make a coordinator fault-tolerant, but which global behaviours need none.

2.2 Named communication: tuple spaces, actors, pub/sub, gossip

Linda's tuple spaces decouple senders from receivers: writers publish, receivers select by template (Carriero & Gelernter 1989); the actor model locates meaning in the receiver's handling (Agha 1986); publish–subscribe and epidemic protocols (Demers et al. 1987) push receiver-side selection to Internet scale. These families anticipate our first principle, but the medium carries *identity*, not *concentration*: a message is delivered or not, it does not accumulate into a graded, decaying field, and nothing in the medium encodes demand.

2.3 Stigmergy and quorum sensing

Stigmergy — ants laying and following evaporating pheromone trails — is the closest biological precedent in computing, formalised as indirect coordination through environment modification (Parunak 1997; Dorigo et al. 1996). Bacterial quorum sensing is its concentration-threshold cousin: cells secrete an autoinducer and switch behaviour above a threshold (Miller & Bassler 2001). Both give concentration semantics with decay, but the repertoire answers "how many of us are here?", not "who needs help, and is supply keeping up?" — and the limits of that repertoire are now measured directly, not merely asserted, in high-density pheromone-swarm robotics (Hunt et al. 2019).

2.4 Amorphous computing and synthetic pattern formation

Amorphous computing asked exactly our question for programmable matter: what global shapes can unreliable, identical, locally-communicating particles achieve (Abelson et al. 2000)? Gradient and threshold primitives were compiled into global behaviours (Nagpal 2003), and synthetic biology realised them in living cells — pattern formation from diffusing signals (Basu et al. 2005), population control with an

Table 1. Coverage of the five coordination primitives P1–P5 and the architectural assumption A1 (Section 3.3) by prior families. ✓ = native; ∼ = partial or incidental; – = absent. The final row is definitional — an internal consistency check that the primitive set describes the reference system, not an empirical comparison.

Family	P1	P2	P3	P4	P5	A1
Central orchestration	–	–	–	∼	∼	✓
Tuple spaces / actors	✓	–	–	–	–	–
Pub–sub / gossip	✓	∼	–	–	∼	–
Stigmergy (ants)	✓	✓	–	–	∼	✓
Quorum sensing	✓	✓	–	–	–	✓
Amorphous computing	∼	✓	–	–	∼	–
Synthetic biology	✓	✓	–	✓	∼	✓
Field-based coordination	✓	✓	∼	–	∼	–
Digital Hormone Model	✓	✓	–	–	∼	–
Artificial endocrine systems	✓	∼	∼	–	∼	–
Artificial Hormone System	✓	∼	∼	–	∼	–
Danger theory / DCA	✓	∼	–	✓	–	–
Congestion / back-pressure	–	∼	∼	–	∼	∼
Antithetic integral feedback	–	∼	∼	–	✓	–
Token-bucket / credit control	–	∼	∼	–	∼	∼
Metabolic / homeostatic	–	∼	∼	–	∼	–
Wound-healing class	✓	✓	✓	✓	✓	✓

explicit kill term (You et al. 2004). The line proves receiver-side threshold logic on a diffusing medium can compute global structure, but stops short of logistics: nothing ramps production by unmet demand, and termination is programmed, not emergent.

2.5 Field-based and hormone-inspired coordination

The closest prior art takes hormones seriously. Field-based coordination computes distributed gradient fields and has agents follow them (Mamei & Zambonelli 2006). The Digital Hormone Model controls modular-robot self-organisation with hormone-like signals that diffuse, carry concentration and decay — the closest precedent to our P1 and P2 (Shen et al. 2004). The artificial-endocrine-system family develops lattice models of endocrine regulation for robotic swarms, requiring no unique identifiers or global information (Xu & Wang 2011), and the Artificial Hormone System assigns real-time tasks using locally computed "hormones" that rise with waiting tasks and fall with load (Brinkschulte et al. 2008). The distinction lies at P3: in these systems the hormone is a *score*, computed and consumed by nodes to guide movement or assignment; no signal is constitutively produced and cleared by its own consumers, so no residual encodes unmet demand, and antagonist dissolution is not a termination primitive. Immune-inspired systems contribute the other half of our safety story: the danger model (Matzinger 1994, 2002) and its dendritic-cell implementation (Greensmith & Aickelin 2005) demonstrate co-stimulus gating — but as classifiers, not coordinators.

2.6 The gap

Table 1 arranges the sixteen families against the primitives defined below. Every family covers a proper subset; no family implements P3, the residual-signal demand loop, natively. Four near-misses demand honesty. Congestion control and back-pressure (Kelly et al. 1998) signal demand through queue backlog — a genuine residual — but the backlog is measured state reported to a computed controller, not a constitutive blind-broadcast signal consumed by its own responders. Antithetic integral feedback (Briat et al. 2016) realises the unpaired remainder of two annihilating species as an error signal — inside a single well, not as a coordination medium across a fleet; it has since been realised experimentally in living and cell-free systems (Aoki et al. 2019), a residue-as-error implementation that remains within a single controlled volume. Token-bucket and credit-based flow control carry a depletion semantics — an exhausted bucket is an availability signal — but per flow, as a rate-policing device, with no loop from depletion to capacity creation. Metabolic and homeostatic regulation lets end-product pools modulate pathway production — conceptually the nearest cousin — but intracellularly, without a broadcast medium or receiver-defined semantics. Beyond the table, the sweep also covered chemical-reaction-network control, demand-driven autoscaling, swarm resource recruitment, and distributed load balancing keyed on availability signals; none couples a constitutive broadcast to consumption-proportional clearance driving capacity creation. The sweep was repeated for the post-2016 threads — event-driven autoscaling in the KEDA style, digital-pheromone swarm robotics, endocrine multi-robot control, and the antithetic-controller line since Briat et al. (2016) (Aoki et al. 2019; Hunt et al. 2019) — with the same result; the protocol, queries, dates and per-thread dispositions are recorded in Appendix A. The families were assembled from the survey literature cited above and snowballed through their taxonomies, each candidate scored against the P1–P5 definitions — fixed before scoring began — by the criteria in Table 1's caption, so the coverage can be audited rather than trusted; an independent second scoring pass is committed before submission (Appendix A). The prior-art test must therefore answer not "has anyone called a signal hormonal?" but: *has anyone built a system in which a constitutively generated quantity is removed in proportion to available service capacity, so that its unconsumed remainder functions directly as the distributed error signal driving capacity creation?* We have not identified such an architecture; the claim is stated as assembled evidence, not proven absence, and it survives as the combination — residual control with receiver-defined semantics on a spatial, decaying medium with antagonists — which is what the programme of Section 4 tests.

3 THE WOUND-HEALING REFERENCE SYSTEM

3.1 The protocol

Fig. 2 walks the repair sequence as a protocol execution. *Haemostasis*: exposed tissue factor initiates the thrombin cascade; thrombin polymerises fibrin and traps red cells into a plug — local chemistry; the "decision" to clot is made by the medium's own state. *Inflammation*: stressed cells emit danger signals — IL-1, complement fragments C5a, CXCL8 — recruiting first neutrophils (phagocytosis) and then macrophages, which clear debris and switch from inflammatory to reparative phenotype as the signal mix changes (Medzhitov 2008; Gurtner et al. 2008). *Proliferation*: growth factors from platelets and macrophages (PDGF, VEGF, TGF-β) drive matrix deposition, angiogenesis, and re-epithelialisation. *Remodelling*: metalloproteinases dissolve the temporary structure over weeks (Gurtner et al. 2008). Handoffs are emergent: each phase's outputs are the next phase's triggers.

3.2 Escalation: the residual-signal demand loop

The load-bearing mechanism is the least discussed, and we present it at three levels: biological mechanism, idealised control law, engineering principle. *Mechanism*: thrombopoietin (TPO) is secreted constitutively, chiefly by the liver; circulating platelets and their marrow progenitors carry the Mpl receptor and clear the hormone from plasma (Kaushansky 2005). When a wound consumes platelets, fewer receptors remain to clear TPO, the residual rises, and the marrow ramps megakaryopoiesis until consumption again matches supply; G-CSF runs an analogous programme for neutrophils under severe demand (Manz & Boettcher 2014). Both systems are richer than any single loop: TPO clearance is coupled to platelet and megakaryocyte mass and hepatic output is itself modulated by further physiological cues, so the hormone is not a purely passive residual detector; and emergency granulopoiesis is not merely "G-CSF doing the same" — it layers additional regulation on the G-CSF axis. That richness is exactly why we abstract.

Control law. Consider the minimal closed loop: a factory secreting a constitutive signal; a circulating pool of responders clearing that signal through receptors, in proportion to their own availability; a workload consuming responders; production of new responders driven by the signal, maturing after delay τ; and a production-relaxation term. With u the responder production rate at baseline u_0, N the responder pool lost at rate μ and consumed by workload $w(t)$, $c(t)$ the receptor-mediated clearance of a constitutive signal flux p_0, and r the residual,

$$\dot{u}(t) = \gamma\, r(t) - \beta\,(u(t) - u_0), \qquad r(t) = [\,p_0 - c(t)\,]_+, \tag{1}$$

$$\dot{N}(t) = u(t-\tau) - \mu N(t) - w(t)\,N(t), \quad c(t) = \kappa N(t), \tag{2}$$

where $[\,x\,]_+ = \max(x, 0)$. Two properties are structural rather than clipped. First, the signal's production p_0 is constitutive — not coupled to factory output; the loop closes through the responder pool, not through the signal, so the causality is depletion $\rightarrow$ residual $\rightarrow$ production. Second, the residual's non-negativity is enforced by receptor saturation: clearance is proportional to responder availability, so depletion — not instruction — raises the residual, and clearance cannot exceed what is circulating. The factory integrates the residual: a workload depletes N, clearance falls, r rises, and production follows — without any agent measuring demand, computing an error, or filing a report. At steady state the residual settles at exactly the level that sustains the production the workload requires ($r^* > 0$ under sustained demand; $r \rightarrow 0$ only when demand vanishes) — supply tracks demand in the sense of matching it, not of zeroing the error. *We are not claiming that Eqs. 1–2 model wound healing.* They are a deliberately reduced control motif extracted from one component of the richer regulation described above — a model of the abstraction, not of the biology. The biological claim retained is deliberately minimal: responder-dependent clearance of a constitutively supplied signal can encode information about responder availability. Nothing about physiological equivalence, organ roles or set-points is claimed, and none is needed for the engineering argument. The decay term is not optional: with $\beta = 0$ the law is pure integral control and u is monotone non-decreasing, so after a transient production is frozen at whatever level it reached — precisely the fibrotic failure mode of Section 3.4. Return to homeostasis requires the antagonist: P3 supplies demand adaptation, P5 supplies reversibility, and the biology implements both, through clearance kinetics and explicit antagonists. Coupling fast concentration dynamics to slow production and maturation makes overshoot and oscillation the expected transient — the biology's own oscillations under chronic demand (Manz & Boettcher 2014) are the model's stress test and Phase I's first theorem. Fig. 3 integrates Eqs. 1–2 through a demand pulse: with the antagonist ($\beta > 0$) the pool recovers and production returns to baseline; with $\beta = 0$ the same pulse leaves both permanently elevated — shown, not merely predicted, for the paper's central law.

Stability. One simulation is not an analysis, and the loop's equilibrium and stability structure can be stated already. Under constant demand w, writing $m = \mu + w$ for the total per-capita loss, Eqs. 1–2 admit a unique positive equilibrium. With the residual inactive ($\kappa u_0/m \geq p_0$), $(u^*, N^*) = (u_0, u_0/m)$ — at zero demand $m = \mu$ and this is the baseline u_0/μ; otherwise

$$N^* = \frac{\gamma p_0 + \beta u_0}{\gamma\kappa + \beta m}, \qquad r^* = p_0 - \kappa N^* > 0, \qquad u^* = m\,N^* : \tag{3}$$

sustained demand settles the pool strictly between the fixed-production ceiling u_0/m (which N^* exceeds exactly when the residual is active) and the zero-demand baseline u_0/μ (which it undercuts whenever $\mu p_0 < \kappa u_0$), with the residual permanently elevated — the loop's normal operating point, not a pathology. Linearising the active branch gives the characteristic equation

$$(\lambda + \beta)(\lambda + m) + \gamma\kappa\, e^{-\lambda\tau} = 0. \tag{4}$$

At $\tau = 0$ every coefficient is positive and the loop is unconditionally stable. For $\tau > 0$, putting $\lambda = i\omega$ on the stability boundary and eliminating the phase $\omega\tau$ yields $(\gamma\kappa)^2 = (\omega^2 - \beta m)^2 + (\beta + m)^2\omega^2$, whose right-hand side is minimised at $\omega = 0$ with value $(\beta m)^2$: a crossing exists *if and only if* $\gamma\kappa > \beta(\mu + w)$. Below that threshold the loop is stable at every delay; above it a Hopf boundary τ_H appears — oscillation is a property of excessive loop gain relative to the antagonist, which delay exposes and antagonist size prevents. The inactive branch is linear with roots $\lambda \in \{-\beta, -m\}$, unconditionally stable. Fig. 3's parameters must be read against both branches. At baseline ($w = 0$, $m = \mu$) the loop sits just above threshold ($\gamma\kappa = 1.5 > \beta m = 1$) at a delay below τ_H, which is why panel (a) shows damped overshoot; during the pulse ($w = 2$, $m = 3$) the inequality reverses ($\beta m = 3 > \gamma\kappa = 1.5$), so the loop is below threshold at every delay — demand raises the per-capita loss that stabilises the loop, making it hardest to destabilise exactly when it is working hardest. Pushing baseline τ past τ_H yields the sustained oscillation Phase II stresses. Phase I completes this into a phase diagram over ($\beta\tau$, $\gamma\kappa/\beta(\mu + w)$) and extends it to finite production capac-

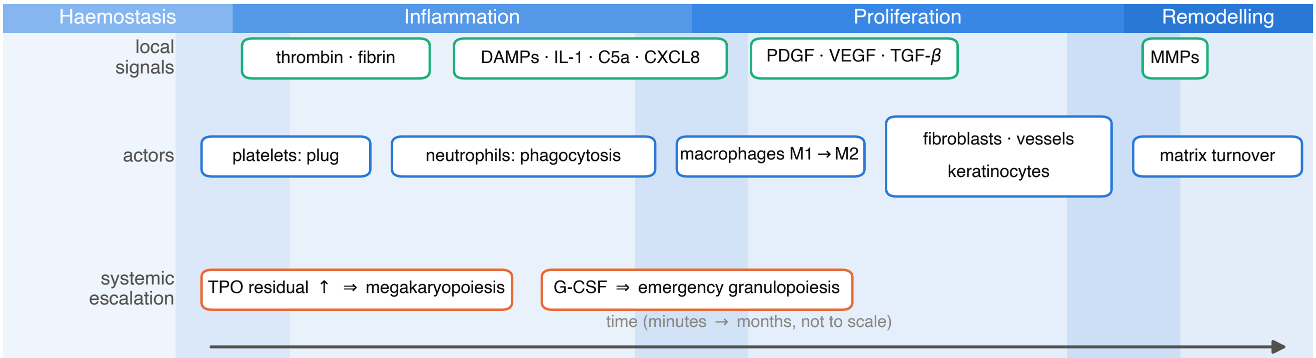

Figure 2. The wound-healing response as a phased protocol with three signal classes. Local signals (aqua) act only on cells carrying the matching receptor; actors (blue) succeed one another because each phase's output is the next phase's trigger; systemic escalation (orange) couples local demand to production of new responders via the residual loop of Eq. 1. Phase bands are drawn overlapping, as in vivo: there is no master state machine declaring a phase boundary — transitions are population-dynamic. No cell holds a global plan; timescales indicative.

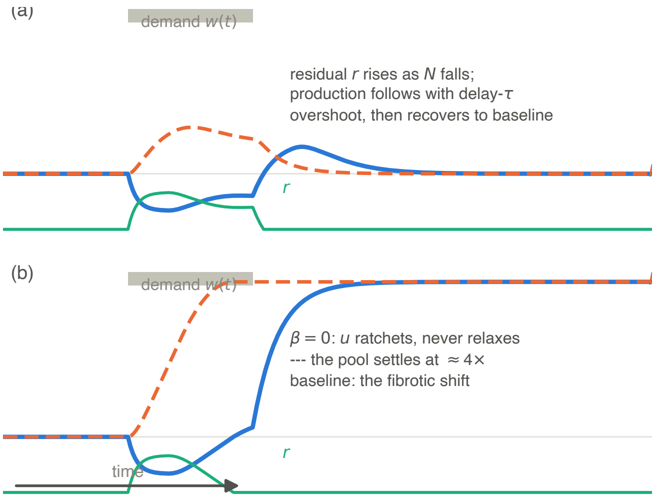

Figure 3. The closed loop of Eqs. 1–2 through a demand pulse ($\mu = \kappa = p_0 = u_0 = N_0 = 1$, $\gamma = 1.5$, $\tau = 1.5$; per-capita demand $w = 2$ on $t \in [5, 10)$). (a) With the antagonist ($\beta > 0$): the residual r (aqua) rises as the pool N (blue) falls; production u (orange) follows with delay-induced overshoot; both return to baseline. (b) $\beta = 0$: production ratchets and never relaxes; the pool settles permanently high — the fibrotic shift. The antagonist term is mathematically essential, not ornamental.

ity, saturating (Michaelis–Menten) clearance, noisy and delayed residual measurement, and heterogeneous responders.

Engineering principle. In control-theoretic terms, Eq. 1 is a leaky integral controller with anti-windup, acting on a non-negative, saturating error through a delayed plant (Åström & Murray 2008) — a known object, and we claim no novelty for the law itself. The novelty is where the error physically resides. Feedback control of computing systems already uses integral action (Hellerstein et al. 2004): an autoscaler integrates a *computed* error from a metrics pipeline. The distinction kept here is where the error lives: measured and integrated by a controller, or *being* the unconsumed remainder of a consumed signal, with the medium doing the computing. This law is what no family in Table 1 implements, and it grounds the programme's deepest hypothesis: *a distributed system can estimate unmet global demand without explicitly measuring global demand, because the residual of a consumed signal can itself encode the error.*

3.3 Five primitives and one architectural assumption

From the protocol and the loop we distil five coordination primitives (Table 2 completes each with its engineering pattern and falsification test):

P1 — Receiver-defined meaning. The same molecule is a growth command to one cell and noise to its neighbour; meaning lives in the receiver's receptors, not in the message.

P2 — Concentration-as-instruction. Signals act by local concentration crossing thresholds, over a medium that diffuses and decays; instructions are graded and spatially local by physics.

P3 — Residual-signal-as-demand. Eq. 1: the unconsumed remainder of a constitutive signal *is* the demand estimate.

P4 — Co-stimulus locality guards. Effector responses require a second, local co-signal (damage-associated patterns, co-stimulation); a solitary systemic signal is not sufficient.

P5 — Antagonist loops and dissolution. Every build-up programme has an antagonist — the β term of Eq. 1 made general — so structures and responses have deadlines.

One assumption is conceptually different and we separate it: A1 — *cheap items, elastic factories.* Coordinated workers are expendable and numerous, and production of new workers is elastic and ignorant — the marrow does not know where the wound is; self-replication is an alternative implementation. A1 is an architectural bet about the substrate, not a coordination primitive. Whether P1–P5 can coordinate expensive, heterogeneous or irreplaceable responders, or silently rely on elastic supply, is an open question the programme tests (Phases II and V) rather than assumes.

Table 2. The validation chain: each primitive at its biological, abstract and engineering levels, with the result that would falsify the transfer (Phase II predictions in parentheses).

	Biology	Abstraction	Engineering	Refuted if
P1	thrombin–PAR; TPO–Mpl	receptor set Σ_a	handlers	addressed messaging loses nothing
P2	VEGF gradient; quorum	threshold on decaying field	TTL counters	no-decay medium performs as well
P3	TPO residual → marrow ramp	Eq. 1	autoscaler on residual	no penalty without it (1)
P4	DAMP + co-stimulation	two-signal AND gate	two-factor actuation	no FPR cut (3)
P5	MMPs; IL-10	decay term β	TTL-based GC	$\beta = 0$ still terminates (2)

One disanalogy must be respected from the start. In biology, receptor identity is fixed by the genome — a molecule either binds Mpl or does not; an adversary cannot forge a receptor. In software the receiver's "receptor" — its subscription set — is dynamic data and can be spoofed: an adversary emitting a stress key with a chosen identity can recruit the fleet against itself. Authentication and authorisation are therefore load-bearing parts of the medium, and P4's co-stimulus guards — a second, locally verifiable signal before effector action — are the architectural mitigation. Both are attacked in Phase V. The mitigation is layered rather than promissory: authenticated emission and key-scoped receptors fix identity — the analogue of genotype-fixed binding; P2's saturation doubles as an innate rate limit, so flooding has bounded effect; P4 requires a second, locally verifiable co-signal, so a spoofed systemic signal alone cannot trigger effector action; and receptor-level anomaly thresholds bound what a valid-but-malicious insider can recruit — the sepsis analogue, which Phase V attacks deliberately precisely because valid keys defeat authentication alone.

3.4 Pathologies as safeguards

The failure modes are as instructive as the successes. Each pathology illustrates the type of failure that arises when one or more of the corresponding control functions become inadequate — a diagnostic mapping, not a claim of single causation, these diseases being multicausal: cytokine storm is dominated by P2 runaway (Tisoncik et al. 2012), answered by saturation, rate limits and circuit breakers; thrombosis is P1/P2 without P4's locality guard, answered by mandatory co-stimulus checks; autoimmunity is miscalibrated meaning, answered by burn-in periods in which response rules are learned or vetoed; fibrosis centrally involves missing P5 — Eq. 1 with $\beta = 0$, demonstrated in Fig. 3(b) — answered by hard dissolution deadlines. Table 2 makes each transfer falsifiable; Phase V makes the pathologies required test chapters. A system that does not ship the antagonists of its own signals has replicated the pathology risk and none of the remedy.

4 A SCIENTIFIC PROGRAMME

The phases are not a waterfall: the minimal loop is simulated as soon as it is written (Fig. 3), control analysis feeds revised models, sandbox anomalies feed back into theorems, and each phase carries a stated negative-result criterion. The criteria are of two kinds, and we label each where it is stated: *termination criteria* — a result that ends the programme (Phase I's stability gate, and Phase II's prediction 1, both of which kill H1) — and *claim-reduction criteria*, where a negative result shrinks the claim rather than ending it: from "inevitable" to "useful" (Phase III), from open fleets to closed fleets (Phase IV), from universal to a mapped envelope (Phase V). The recurring scientific object is delay: fast concentration coupled to slow production, the likely source of overshoot and oscillation.

The programme tests three hypotheses that can fail independently. *H1, mechanistic*: residual consumption is sufficient to estimate unmet demand — Eqs. 1–2 deliver stable, reversible demand tracking. *H2, architectural*: the five primitives together produce robust coordinator-free demand covering that no subset reproduces. *H3, comparative*: over a defined operating envelope, the architecture holds a better cost/robustness frontier than conventional alternatives at equal information and resource budgets. H1 is tested by Phase I and prediction 1; H2 by the Phase II factorial (predictions 1–3); H3 by Phases III–IV (prediction 4). A negative H1 terminates the programme; H2 and H3 can each fail without taking H1 down, and Table 3 maps every phase outcome to its response so the falsifiability claims can be audited rather than admired. The programme also carries two separable novelties that should not be conflated: a *scientific* claim — residual-demand signalling is a stable, discoverable control mechanism (H1, H2) — and an *architectural* claim — it is competitive on real distributed substrates (H3). H1 can succeed while H3 fails, and the result would still be interesting: a validated, biology-derived control law that conventional infrastructure happens to beat on its own turf is a finding about the biology of control, not a failure of the science.

4.1 Phase I — Minimal abstraction and controller analysis

Define a hormonal system HS $= \langle A, \Sigma, M, R, D\rangle$: agents A; a signal alphabet Σ; a medium M carrying concentration fields; threshold-gated response functions R; and degradation/clearance D. Two signal classes are distinguished explicitly, carrying different physics and cost. *Local fields* serve spatial recruitment,

$$\frac{\partial m_s}{\partial t} = D_s \nabla^2 m_s - \lambda_s m_s + \sum_{a \in A} e_{a,s}(t)\, \delta(\mathbf{x} - \mathbf{x}_a), \quad (5)$$

while *well-mixed systemic signals* serve supply loops: their medium is a circulating pool rather than a spatial field — TPO is whole-body — and their dynamics are the ODEs of Eqs. 1–2. The phase's first deliverable is that closed loop itself — the smallest system containing factory, pool, workload, clearance and delay — analysed before the full HS. Each agent acts on local readings alone:

$$u_a(t) = F_a(\mathbf{1}[m_s(\mathbf{x}_a, t) \geq \theta_{a,s}] \text{ for } s \in \Sigma_a,\ \text{local state})\,. \quad (6)$$

The committed task class is deadline-constrained demand

Table 3. Decision table: how phase outcomes map to programme responses. "Negative" is the phase's stated criterion firing — termination for I and II, claim reduction for III–V; combinations not shown are revisions — sandbox anomalies feed back into Phase I models by design.

Phase	If negative	If positive
I	no stable region at realistic τ: *terminate*	stability map secured: proceed
II	no P3 penalty: *H1 fails; programme ends as stated*	factorial structure confirmed: H2 stands
III	no rediscovery: H1, H2 unaffected; claim narrows from "inevitable" to "useful"	convergent rediscovery: strongest result
IV	substrate cost dominates: H3 fails for open fleets; scope to closed fleets	parity under denial: H3 stands
V	envelope too narrow: claims scoped to the mapped envelope	broad envelope: an engineering statement

covering: a set of sites each requires r_i responders within time T of demand onset, under churn. The target theorems, with named techniques: (i) existence and uniqueness of the equilibria of Eqs. 1–2, their stability, overshoot bounds and the delay-Hopf boundary of Eq. 4 — begun in Section 3.2 (unconditional stability at $\tau = 0$; the delay threshold $\gamma\kappa$ versus $\beta(\mu + w)$) — completed as a phase diagram over $(\beta\tau,\ \gamma\kappa/\beta(\mu+w))$ by linearisation, Routh–Hurwitz and Lyapunov/contraction arguments, and extended to finite production capacity, saturating (Michaelis–Menten) clearance, noisy and delayed residual measurement, and heterogeneous responders; "realistic" τ being anchored in the reference biology: hormone half-lives of order hours against platelet lifespans of 7–10 days and multi-day granulopoiesis give $\tau_{\mathrm{production}}/\tau_{\mathrm{signal}} \sim$ 10–100, and the theorems must hold across that range; (ii) a message bound of $O(n\,k)$ for this class, by potential-function argument, with the medium's broadcast accounting as $O(1)$ per emission; (iii) termination under antagonist completeness (every generated structure has a dissolution rule); and (iv) an impossibility boundary in the spirit of Fischer et al. (1985): which specifications provably cannot be met by receiver-only semantics, where coordinators remain necessary.

The phase's deliverable is a volume, not a tuned point: what fraction of the anchored parameter box — τ spanning the reference ratio above, $\gamma, \beta, \kappa, \mu, w$ anchored to the ranges the biology and the Phase IV substrates motivate — is asymptotically stable under the quantitative bounds, and how much of that volume survives a simultaneous ±20% error in every parameter. A mechanism that needs four parameters tuned to a few per cent is not a candidate for robust distributed computing, so the strongest possible Phase I result is a large stable volume — majority, not sliver — retained under parameter uncertainty, with Sobol indices naming which parameters the envelope is most sensitive to and which are safe to leave unidentified.

The complexity claim is carried by a five-component vector $(C_{\mathrm{agent}}, C_{\mathrm{medium}}, S_{\mathrm{agent}}, S_{\mathrm{medium}}, L)$: per-agent communication (agent-to-agent messages), medium communication (emissions, deliveries, bytes, replication, medium CPU and network work), per-agent state, medium state, and latency (energy joins L when Phase IV reaches edge hardware). A well-mixed broadcast reaches all n nodes: its delivery cost is C_{medium}, not zero. A coordinator hidden inside the medium is still a coordinator — theorem (ii) is honest only if every component is bounded, and this vector, not a scalar message count, is the programme's cost statement throughout.

Termination criterion: if no parameter region with realistic production delay τ — $\tau_{\mathrm{production}}/\tau_{\mathrm{signal}} \sim$ 10–100 — leaves Eq. 4 asymptotically stable with overshoot $\leq 50\%$ and settling within five production delays (the same bounds Phase V enforces, so "stable" is quantitative: a tolerance, not a vibe), the load-bearing loop is unsound and the programme ends.

4.2 Phase II — Digital wound sandbox

An executable testbed implementing Eqs. 5–6 on discrete topologies, seeded with three scenario classes translated from biology: *injury* (node loss or surge), *infection* (sustained load requiring supply expansion), and *chronic demand* (persistent stress testing termination). Every run instruments the full causal chain — demand → residual → production → population → recovery. The design instrument is ablation in two layers: single-principle knockouts, and a factorial design over P1–P5 with interaction terms, because the primitives' value may be conditional on each other — P3×P5 (integration without antagonist: oversupply and oscillation), P2 without P4 (false activation), P1+P2 without damping (unstable propagation), P4 without P2 sensitivity (failure to respond). Methodology: ≥ 30 seeds per cell, bootstrap 95% confidence intervals, and effect sizes preregistered once the noise and adversarial models are fixed, over a fractional replicate of the 2^5 design ranked by first-order Sobol sensitivity indices computed from the Phase I linearisation (provisionally 10^5 simulated node-hours per scenario class). Beyond the scenario classes, a dedicated perturbation-and-recovery battery — step, impulse, ramp, periodic and stochastic demand; abrupt responder depletion and restoration; signal delay and loss; heterogeneous receptor thresholds — measures rise time, settling time, overshoot, integrated error, oscillation amplitude, resource excess, communication cost and failure probability, making the loop a distributed-control experiment whether or not the biological analogy is credited. P3-removal is not the only decisive arm: the decisive comparison runs six arms — full P1–P5; P3 removed; ordinary measured-error integral feedback granted identical information; queue/back-pressure control; decentralised gossip load-balancing; and a central oracle — answering whether residual signalling merely works, or confers an identifiable advantage over conventional feedback at equal information and resources. The predictions the thesis stands or falls on:

(H1) The principal comparison is the outcome vector — recovery time, integrated error, overshoot, resource excess, communication cost, failure probability — across the six arms at matched information and budget; the scaling exponent below is one derived component of it, not the whole test. Without P3, production is fixed and the pool is capped at the ceiling u_0/m, so time-to-resolution $T = -(1/m)\ln[(u_0/m - N_T)/(u_0/m - N_0)]$ grows superlinearly in burst size and diverges as the requirement N_T approaches that ceiling, while the full model's demand-tracking production keeps T near-linear. Phase I derives the divergence point and the near-ceiling form; the numeric threshold is preregistered before the definitive runs.

(H2) Disabling P5 (antagonists) yields runaway accumulation — the fibrosis signature — in a measurable fraction of chronic-demand runs.

(H2) P4's co-stimulus gate strictly reduces false recruitment ($\mathrm{FPR}_{\mathrm{P4}} < \mathrm{FPR}_{\text{no-P4}}$) under adversarial signal injection; the effect size is preregistered, not assumed.

(H3) Under churn and coordinator-targeted denial, a hormonal fleet degrades gracefully where a Kubernetes-style orchestrated baseline of equal steady-state throughput fails outright (A1 alone does not save it).

Termination criterion: if the full model loses to conventional feedback on the preregistered outcome vector at matched information and budget — no penalty for removing P3, no advantage to keeping it — the residual loop is not the mechanism the thesis rests on, and H1 fails. A laptop-scale pilot implementation of both phases — code, seeds and raw outputs in the repository (`Jay/phase12`) — is reported in Appendix B; it was run to de-risk the two termination criteria ahead of the definitive runs, and its findings are marked where they sharpen the commitments made here.

4.3 Phase III — Benchmarks and inverse design

Hormonal coordination earns its place only against strong alternatives, in five tiers under identical workloads, information and resource budgets: a centralised oracle and a near-oracle; a conventional distributed controller; a state-of-the-art autoscaler (Kubernetes-style orchestration); the closest field/hormone systems (gossip membership, stigmergy-style field following); and our own ablated variants. A comparison against handicapped baselines proves nothing. The oracle tier is an implementation convenience, not an architecture: a centralised scheduler granted simulated direct access to global state at equal compute and message budget — an upper bound on what any coordinator could achieve — while the near-oracle adds realistic measurement latency; all tiers are run at budgets the phase polices explicitly. Hand-designing receptor sets will not scale; the amorphous-computing lesson is to *compile* global shape into local rules (Nagpal 2003), and the synthetic-biology lesson is that search finds rule sets analysis cannot (Basu et al. 2005; You et al. 2004). The programme therefore includes evolutionary search over an extended genotype — (Σ, R, θ) plus medium parameters $(\lambda, D, \gamma, \beta)$ and affinity/antagonist relationships — scored against two target specifications (deadline covering; graceful degradation under churn) under a fixed compute budget (provisionally 10^6 evaluations per specification). The joint space is not searched blind: a pilot — fixed topology, two or three loci free at a time — validates the fitness landscape before the extended genotype is released. Search output is not trusted for fitness alone: high-fitness solutions are audited against Phase I's preconditions and inspected for which primitives they use.

The audit carries the scientific payload and the programme's central evolutionary hypothesis: *deprived of global state but required to solve variable-demand resource allocation robustly, optimisation independently evolves residual-demand signalling.* The test is run on a ladder of progressively stronger global-information restrictions, so the motif's emergence — if it emerges — can be attributed to the restriction and not the fitness function. The search is also blind, with the blindness frozen in advance: the genotype, the fitness function, the information budget, the criteria for judging a loop P3-like, and the classification procedure are all fixed and dated before the first run of the released genotype; the genotype carries no signal semantics, the fitness function never names primitives, and evolved architectures are classified only after the runs complete — de-identified, with condition labels unblinded after every solution is classified — so rediscovery is convergent evidence, not scaffolding. The joint space is of order 10^2 dimensions (eight signal loci with emission, response and threshold genes each, four medium parameters, an 8×8 affinity/antagonist matrix), which is why the pilot measures the landscape's correlation length before the full release. Does search rediscover the biological motifs — co-stimulus gating, antagonist pairing, constitutive-and-consumed signals — without being told to? If audited solutions converge on P3-like loops across specifications, the primitives are a general control architecture rather than biology trivia — evidence of convergent discovery by biological evolution and computational optimisation, and the programme's most impressive possible result.

Claim-reduction criterion: if audited solutions that meet specification never contain residual-demand loops, P3 is not a discoverable general mechanism — the claim narrows from "a general control architecture" to "a hand-designed one"; H1 and H2 are unaffected.

4.4 Phase IV — Substrate engineering

Deployment needs a medium with the physics of Eq. 5, which present-day infrastructure approximates: publish–subscribe with per-key time-to-live aggregation windows provides diffusion-and-decay counters; sidecars provide the receptor abstraction over legacy services; conflict-free replicated data types (Shapiro et al. 2011) give idempotent resumption when responders die mid-task. Before either pilot, a medium audit connects Eq. 5 to the discrete substrate: emulation on a grid testbed measures the approximation error of TTL aggregation against the continuous field, and the pilots proceed only if that error is bounded — Phase I's guarantees must survive the discretisation they are deployed on.

The medium's own failure modes are part of the architecture, not an afterthought. Under partition, concentration fields decay in place, and staleness is self-announcing: a decaying signal reads as *less* demand, so a partitioned region under-responds rather than mis-responds, with local baseline capability carrying the interval. Concentration semantics also relax delivery requirements: aggregation is commutative and idempotent, so at-least-once delivery with per-key time-to-live is sufficient, message ordering is irrelevant, and exactly-once semantics — so expensive in messaging systems — are simply not needed. Sustained medium loss remains fatal to escalation (the residual cannot be transported), which is why medium availability and partition behaviour are axes of the Phase V boundary rather than footnotes.

Two pilots, each with success criteria stated against existing practice: (a) microservice incident self-management: a stressed service emits a stress key, co-stimulus-gated responders drain queues and shed load, and an autoscaler keyed on the *residual* of a constitutive heartbeat replaces metrics pipelines — the criterion is parity with a Kubernetes horizontal autoscaler and an event-driven autoscaler in the KEDA style on benign step loads, responding no slower than one metrics interval, under identical replica counts, network allocation and response capacity for both sides, while surviving coordinator-plane denial that kills both and riding through an injected 30-second medium partition with no erroneous action (fields decay in place, behaviour falls back to local baseline capability, recovery proceeds without thrash); (b) edge/swarm task allocation, an AHS-scale problem (Brinkschulte et al. 2008) extended with supply dynamics — the criterion is throughput within 10% of a centralised assigner at 10× the fleet size, at bounded C_{medium}, under the same medium-partition ride-through criterion.

Claim-reduction criterion: if the medium's substrate cost dominates the coordination saved on realistic fleets, the architecture is a renaming of the coordinator — H3 fails for open fleets, and the claim is scoped to closed fleets rather than abandoned.

4.5 Phase V — Failure theory and the operating boundary

Section 3.4's four pathologies become required test chapters, each with quantitative damping bounds rather than happy-path demonstrations: signal spoofing (the sepsis analogue — an adversary injecting stress keys to exhaust the factory; mitigated by medium authentication and P4, attacked directly), storm cascades, miscalibration after topology change, and fibrotic accumulation. And the question the existence proof cannot answer must be answered here: *where is the operating boundary?* The programme maps degradation against demand amplitude, spatial extent, latency, noise and responder depletion — severe trauma overwhelms even biology — and delivers the boundary as an operating-envelope map,

$$\begin{aligned} &F(\mathrm{demand}, \tau, \mathrm{noise}, \mathrm{churn}, \mathrm{capacity}, \\ &\quad \mathrm{attack\ rate}, \mathrm{medium\ availability}) \\ &\to \{\mathrm{stable}, \mathrm{underdamped}, \mathrm{oscillatory}, \mathrm{runaway}, \mathrm{collapse}\}, \end{aligned} \tag{7}$$

summarised as phase diagrams — $\tau_{\mathrm{production}}/\tau_{\mathrm{signal}}$ against peak demand over baseline capacity, and $\beta\tau$ against normalised loop gain $\gamma\kappa/\beta(\mu+w)$ from Eq. 4 — turning "severe trauma overwhelms even biology" into a quantitative operating envelope. Provisional targets: post-step overshoot $\leq 50\%$ (a bound matched to typical autoscaler error budgets, and revisited against Phase II data), recovery to within 10% of baseline within five production delays, zero runaway accumulations in 10^4 chronic-demand node-hours — aligned with conventional autoscaling SLOs and error budgets, to be tightened after Phase II. The headline deliverable is a region statement, not a point claim: the envelope map names where hormonal coordination *dominates* (high churn and coordinator-targeted denial, where the medium's cost is amortised and the coordinator is the attackable part), where it is merely *comparable* (benign steady loads, where conventional feedback is good enough), and where it *fails* (sustained partition and extreme latency, where the medium itself is the bottleneck). Only then does "coordination without a coordinator" become an engineering statement with a known domain.

Claim-reduction criterion: an operating envelope too narrow to cover operationally relevant workloads scopes the architectural claim to the mapped envelope — H3 survives inside it and is not asserted outside it.

4.6 Feasibility and resourcing

The programme is sized as a small-team effort of roughly three years, and its budgets are derived rather than asserted. Phase I is analyst work plus laptop-scale numerics ($\sim 10^3$ core-hours). Phase II's 10^5 node-hours follow from the design: ≥ 30 seeds per cell resolves standardised effects $d \gtrsim 0.7$ at 95% confidence with bootstrap intervals, and the budget covers the fractional factorial, the perturbation battery and the six-arm comparison across three topologies. Phase III's 10^6 evaluations run in weeks on a 64-core node once the pilot has validated the fitness landscape; Phase IV requires two engineers and existing cluster hardware; Phase V is analyst and red-team time. The compute figures are initial budgets, and compute is not the binding constraint: Phase III's expensive parts are design, implementation, artefact recognition and validation — building the simulator, the blinding protocol and the classifier, redesigning fitness after the pilot, auditing evolved solutions for hidden global state — which are engineer-months, not core-hours. Every budget is provisional, gated by the preceding phase's pilot, and allocated by the computed Sobol ranking — so a funding panel can audit each number back to a design choice rather than a guess.

5 DISCUSSION

Three clarifications keep the thesis honest. First, the complexity claim is a vector, not a scalar: the agent-side components of $(C_{\mathrm{agent}}, C_{\mathrm{medium}}, S_{\mathrm{agent}}, S_{\mathrm{medium}}, L)$ fall to $O(n\,k)$ conditional on a medium providing broadcast, locality and decay, while what that medium spends is a real cost paid in Phase IV; the medium is the load-bearing wall, and without decay, concentration semantics degenerate and with them P2 and P3. Second, the class of reachable behaviours is restricted to what threshold logic on fields can express; Phase I's impossibility results are expected to show that some specifications (strict global invariants, exact sequencing) genuinely require coordinators. The claim is not that coordinators are never needed; it is that a large, operationally important class — incident response, load balancing, supply tracking, cleanup — does not need them, and pays less for not having them. Third, biology keeps centralised *production* (A1); the argument is against centralised *deciding* — the marrow does not know where the wound is, and does not need to.

The comparison to autonomic computing (Kephart & Chess 2003) is generational: autonomic systems report to a control loop; a hormonal system embeds the control loop in the medium and lets the residual do the reporting. If the programme's predictions hold, the practical outcome is systems whose coordination cost does not compound with scale — where, within the mapped operating envelope, the thousandth node is as cheap to coordinate as the tenth, because it joins a medium rather than a meeting.

6 CONCLUSIONS

We have posed coordinator-free coordination as a precise question, shown that wound healing answers it once in nature over a substantial operating envelope, distilled that answer into five testable coordination primitives plus one architectural assumption, derived the central control law's equilibria and delay-stability threshold and demonstrated its fibrotic failure mode in simulation (Fig. 3), located the gap in the sixteen families surveyed (the residual-signal demand loop above all), and specified a five-phase programme with falsifiable predictions, stated criteria per phase — terminal for the core mechanism, claim-reducing elsewhere — and a decision table — testing three separable hypotheses: mechanistic (H1), architectural (H2) and comparative (H3), whose two novelties are separable: the science of the residual loop can stand even where its engineering case does not. The programme's terminal output is a dominance map — where hormonal coordination wins, where it ties, and where it should not be used. The question the programme answers is the one the introduction asked; the standard of success is the one the reference system already meets: many items, each knowing the bare minimum, doing excellent work together.

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APPENDIX A: SEARCH METHODOLOGY FOR THE GAP ANALYSIS

The gap analysis of Table 1 was run as a falsification directed search: the goal was not to assemble supporting citations but to find an architecture that implements P3 natively and thereby falsify the claimed gap before publication.

Protocol. Sources: Google Scholar, ACM Digital Library, IEEE Xplore, arXiv (cs.MA, cs.DC, eess.SY) and PubMed for the biology-side precedents. Query families: hormone/endocrine regulation for multi-robot and distributed coordination; residual or unconsumed signal as demand estimate; antithetic and sequestration integral feedback; pheromone, digital pheromone and stigmergy at scale; back-pressure and congestion-driven capacity scaling; event-driven and metric-pipelined autoscaling; amorphous and homeostatic computing. Each family's literature was then snowballed through its own taxonomies and reviews. The original survey was assembled 2026-07 and the currency re-sweep of post-2016 threads was run 2026-08-15.

Inclusion and scoring. A candidate enters the table if it claims distributed coordination or resource allocation through signal fields or a chemical metaphor. The P1–P5 and A1 definitions (Section 3.3) were fixed before any family was scored. The rubric: P1 is native when receiver-side selection is load-bearing, partial when addressing dominates; P2 is native when graded, decaying concentration is load-bearing, partial when signals are binary; P3 is native only when a constitutively produced quantity is cleared in proportion to available responding capacity and its residual drives capacity creation — partial when a measured backlog or computed error approximates the residual; P4 native when two-signal gating is required for effector action; P5 native when antagonist dissolution terminates responses.

Dispositions of the post-2016 threads. Antithetic sequestration control has been realised experimentally since Briat et al. (2016), in living and cell-free systems (Aoki et al. 2019), but remains a single-volume controller: no broadcast medium, no receiver-defined semantics — row unchanged. Pheromone stigmergy has been tested to its limits at high swarm density (Hunt et al. 2019); the repertoire is spatial recruitment, with no constitutive-consumed residual — row unchanged. Event-driven autoscaling in the KEDA style scales on queue depth and event backlog — a genuine residual, but one measured and integrated by a controller through a metrics pipeline; its canonical sources are project documentation and vendor literature rather than peer-reviewed evaluations, which we note as a finding about the thread, and it is scored under congestion/back-pressure — row unchanged. Successors of the artificial-endocrine family keep the hormone as a node-computed score; no post-2016 system found couples a constitutive broadcast to consumption-proportional clearance driving capacity creation.

Commitments. The frozen definitions, the query list, the dates and this rubric are preregistered with the Phase I–II registration. Before submission, an independent second scorer rescores every family against the same frozen definitions; disagreements are resolved by documented discussion, and both scorings will be published with the paper.

APPENDIX B: PHASES I–II PILOT: IMPLEMENTATION AND FIRST RESULTS

This appendix reports a pilot execution of the Phase I and Phase II commitments (Sections 4.1–4.2), run on 2026-08-16 to de-risk the programme's two termination criteria before the definitive runs. The pilot implements the closed loop of Eqs. 1–2 as a delay differential equation integrated by Euler steps with a ring-buffer maturation history, and the sandbox of Eqs. 5–6 on a 24×24 torus with positivity-preserving donor-cell advection; P1–P5 are compile-time toggles and the six arms are production/redistribution policies on the same plant. Every draw is through a seeded `numpy` generator; code, seeds and raw outputs are in the repository (`Jay/phase12`: `jay_core.py`, `sandbox.py`, `run_phase1.py`, `run_phase2.py`, `phase1_results.json`, `phase2_results.json`). The entire pilot — all sweeps, both phases — runs in under three minutes on one laptop. It is not the definitive study: 30 seeds per sandbox cell, one grid resolution, and no preregistered effect sizes; its numbers are descriptive. Its purpose is to find, early and cheaply, the places where the programme's stated commitments meet an unfriendly numerical reality; five such findings are collected in Section B3.

B1 Phase I pilot: the minimal loop

Setup. The normalisation $\kappa = p_0 = 1$, $u_0 = \mu$ makes the no-demand baseline $N = 1$ with zero residual, generalising Fig. 3's. The anchored box of Section 4.1 is sampled log-uniformly in its five free dimensions: $\mu \in [10^{-2}, 10^{-1}]$ (responder lifespans), $\beta \in [10^{-2}, 0.3]$ (production relaxation), $\tau \in [10, 100]$ (the reference ratio $\tau_{\text{production}}/\tau_{\text{signal}}$), loop gain $y = \gamma\kappa/\beta(\mu + w) \in [0.5, 10]$ (the ordinate of the Section 4.1 phase diagram, so γ is fixed by the drawn y), and demand $w/\mu \in [1, 10]$. A box point *passes the quantitative gate* if the active branch is asymptotically stable under Eq. 4 *and* the simulated step response overshoots the target by $\leq 50\%$ *and* settles within the $\pm5\%$ band in at most five production delays — the same bounds Phase V enforces.

Equilibria. Across 400 box points, the simulated final state matches Eq. 3 to 0.55% (95th percentile) wherever the response has converged ($\tau < 0.5\,\tau_H$); the error tail (6.7% at the 95th percentile over all stable points) sits entirely at $\tau/\tau_H \in [0.5, 1)$, where damped ringing outlasts the 12τ integration window — slow convergence near the boundary, not integrator bias.

Phase diagram. Fig. B1(a) plots the closed-form Hopf boundary τ_H of Eq. 4 in the $(\beta\tau, y)$ plane at three values of m/β, with the Monte-Carlo box overlaid in the same coordinates. Simulated oscillation onset coincides with the analytic boundary: at the Fig. 3 slice, the last-quarter amplitude is ≤ 0.002 at $0.5\,\tau_H$, jumps $\times4$ across τ_H (0.0125 at $0.8\,\tau_H$ to 0.047 at τ_H at $y = 1.5$) and keeps growing to 0.083 at $2\,\tau_H$; the same stair-step appears at $y = 3$ and $y = 6$.

Gate volume — the honest number. Over 2^{12} box points: 45.9% are asymptotically stable, 36.3% meet the overshoot bound, and 11.5% settle within five production delays — the gate volume is 11.5%, and *settling is the binding criterion* (median settling among passers: $4.17\,\tau$; median overshoot: 10%). Under eight simultaneous $\pm20\%$ perturbations of all five dimensions, 50.3% of passing points survive every draw (5.8% of the box unconditionally). Against Section 4.1's hope of "majority, not sliver", the pilot finds a sliver — and Section B3 records the two candidate responses (denominate the settling budget in the loop's slowest timescale, or report the volume honestly). The termination criterion itself is *not* triggered: a realistic-delay region passing all three bounds with majority robustness exists.

Sensitivity. First-order Sobol indices of the gate indicator (Fig. B1b): w 0.75 [0.65, 0.86], μ 0.71 [0.61, 0.81], τ 0.56 [0.47, 0.68], β 0.40 [0.31, 0.50], and loop gain y 0.04 [−0.02, 0.11] (bootstrap 95% CIs). The inversion is the useful result: which parameters must be identified carefully are the timescales and the demand; the loop gain that the stability analysis centres on is, within the anchored box, nearly free — exactly the parameter the biology leaves unpinned across an order of magnitude. The indices sum to well above one's expectation for an additive decomposition (2.5), i.e. the gate fails through interacting parameter combinations, consistent with settling being governed by the slowest of $\{\tau, \beta^{-1}, m^{-1}\}$.

Extensions. Table B1 gives the gate volume under the four named extensions. Finite capacity and Michaelis–Menten clearance *help* (both self-limit overshoot); heterogeneity costs a third of the volume; and the delayed+noisy measurement arm collapses it — decomposed, the collapse is entirely measurement *latency*, not measurement *noise*: with

Table B1. Phase I gate volume under the four named extensions ($n = 2048$ box points each; baseline re-drawn per arm).

arm	gate volume
baseline	0.123
finite capacity ($u \leq 4u_0$)	0.140
Michaelis–Menten clearance	0.146
heterogeneous responders	0.085
measurement noise only ($\sigma = 0.1$)	0.123
residual delay $0.1\,\tau$ only	0.091
residual delay $0.25\,\tau$ only	0.048
delayed+noisy residual ($0.5\,\tau$, $\sigma = 0.1$)	0.015

Table B2. No-P3 divergence: time to reach $N_T = (1 - \varepsilon)\,u_0/m$ from $N_0 = 0.3$ ($\mu = w = 0.05$, $m = 0.1$, ceiling $u_0/m = 0.5$; tracking loop $y = 0.6$, $\tau = 20$). "Theory" is the closed form of H1 for the fixed arm.

ε	T_{fixed} (sim)	T_{fixed} (theory)	T_{tracking} (sim)
0.355	1.20	1.20	1.20
0.092	14.72	14.73	14.72
0.024	28.24	28.25	21.22
0.0061	41.74	41.78	21.62
0.0016	55.26	55.31	21.72

$\sigma = 0.1$ sampled-and-held jitter and no delay the volume is unchanged (the integral loop averages the jitter), while a pure residual-measurement delay of $0.1\,\tau$, $0.25\,\tau$, $0.5\,\tau$ gives 0.091, 0.048, 0.015. The minimal model's assumption that the medium relaxes on the signal timescale is load-bearing: the factory must see the residual fast relative to its own production delay, or the settling budget is spent before the loop begins.

The no-P3 divergence (closed-loop face of H1). With demand on throughout and a burst removing 70% of the pool ($N_0 = 0.3$ under a fixed-production ceiling $u_0/m = 0.5$), Table B2 compares the simulated time to reach $N_T = (1 - \varepsilon)\,u_0/m$ against the closed form $T = -(1/m)\ln[(u_0/m - N_T)/(u_0/m - N_0)]$. The fixed-production arm matches the closed form to $\leq 0.1\%$ at every ε and diverges logarithmically as $\varepsilon \to 0$, while the demand-tracking loop ($y = 0.6$, delay-unconditionally stable) saturates at $\approx 1.1\,\tau$: at $\varepsilon = 1.6 \times 10^{-3}$ the contrast is $2.6\times$ and growing without bound. Near the ceiling the two arms agree, as they must — the loop has not yet had time to matter.

B2 Phase II pilot: the digital wound sandbox

Setup. Three sites of demand on a 24×24 torus; responders migrate up the sensed gradient above threshold (χ-limited, donor-cell advection) and are expended by serving; the systemic loop of Eqs. 1–2 couples a factory (delay $\tau = 20$, antagonist β) to the grid-total pool through the residual $r = [1 - \kappa N]^+$. Scenario classes: *injury* (one-shot demand), *infection* (demand refills at constant rate), *chronic* (refilling demand that clears at $t = 6\tau$). Routine housekeeping traffic — diffusive distractor pulses that receiver-correct responders must ignore — runs throughout (P1's payload). 30 seeds per cell; every metric below is over seeds; the deadline is 8τ.

Factorial knockout. Table B3 regresses the failure indicator on the five toggles plus the three named interactions (2^5 cells, infection scenario). P3 dominates: knocking out the residual

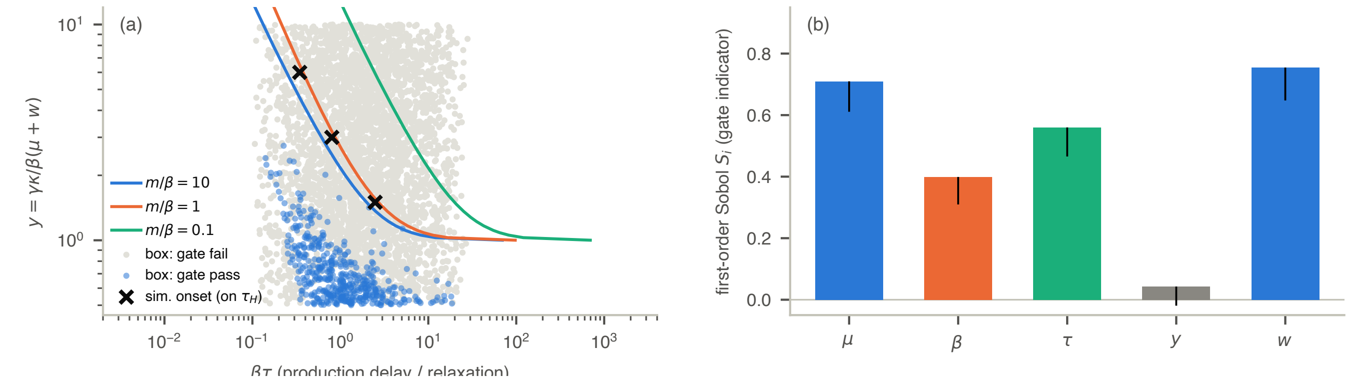

Figure B1. Phase I pilot. (a) The delay-Hopf boundary of Eq. 4 in the $(\beta\tau,\, y)$ plane of Section 4.1 at three m/β contours (lines, left to right: $m/\beta = 10,\ 1,\ 0.1$); the anchored parameter box overlaid as pass (blue) / fail (grey) against the quantitative gate; $\times$ marks simulated oscillation onset, placed at $\tau = \tau_H$. (b) First-order Sobol indices of the gate indicator (covariance estimator, Saltelli A/B matrices, $N = 1024$, bootstrap 95% confidence intervals): the envelope is set by timescales and demand (μ, w, τ, β), not by loop gain y.

loop costs 0.763 $[-0.808,\ -0.708]$ in failure probability — every P3-less cell fails outright under infection, exactly the no-P3 ceiling of H1. P2 and P4 knockouts cost 0.075 and 0.083. The P3$\times$P5 interaction is the largest conditional effect (-0.175): the antagonist's dissolution pays only when there is a loop to terminate — with P3 present, adding P5 cuts failure from 0.24 to 0.07 averaged over the stratum; with P3 absent it is irrelevant. (The P5 main effect is unidentified rather than zero: all sixteen P3-less cells fail at exactly 1.0, so the saturated stratum has no variance and the effect loads entirely on the interaction.) The P2$\times$P4 term (-0.058, CI marginally excluding zero) points the same way as the design intent: the co-signal guard substitutes for some of the selectivity that threshold-free responders lose. P1 is failure-neutral at these distractor loads (its cost appears as misallocated holding, not failed episodes) — an honest null.

Two further readings of the cell means matter for the definitive design. First, in the *benign* infection scenario the P3-only cell fails least of all (0.00 failure, 0.059 unmet, versus 0.10 / 0.077 for the full stack): the guard and the antagonist each trade a little throughput for safety that this scenario does not bill — their value appears precisely where H2 and H2b bill it. Second, the P4 knockout *raises* failure here (0.00 $\to$ 0.33 in the P3-only stratum) because serving gated by the co-signal is slower *and* because without the guard, responders burn at stale halos left by cleared sites — friendly fire, not adversarial fire. The primitives are throughput costs and robustness gains, which is the paper's argument for reporting the full outcome vector rather than latency alone.

Six arms at matched information. Table B4 reports the decisive comparison across the three scenarios, with the medium-side cost vector. The honest ordering on latency: the oracle resolves fastest ($0.31\,\tau$, reading global state); queue/back- pressure and gossip next ($0.9\,\tau$, routing directly on per-cell load); the two residual-driven arms at 3.5–$4.8\,\tau$; fixed production fails outright wherever demand refills. The *measured-error* arm — an ordinary integral controller granted site reports — tracks the physical residual almost exactly (within 2% on resolve time and unmet, infection) at 1.5 messages per unit time: the medium computes the same integral for free, and that equivalence, not superiority, is the correct reading of P3 at this scale. The hormonal arm's case

Table B3. Factorial knockout regression on the infection scenario (2^5 cells $\times$ 30 seeds; toggle coded 1 = knocked out; OLS on cell means, bootstrap-over-seeds 95% CIs).

term	failure	unmet
intercept	$+0.937$ $[0.914, 0.960]$	$+0.190$ $[0.180, 0.199]$
P1	-0.008 $[-0.038, 0.025]$	-0.006 $[-0.014, 0.002]$
P2	$+0.075$ $[0.033, 0.119]$	$+0.005$ $[-0.004, 0.014]$
P3	-0.763 $[-0.808, -0.708]$	-0.109 $[-0.115, -0.102]$
P4	$+0.083$ $[0.050, 0.119]$	-0.011 $[-0.018, -0.004]$
P5	(unidentified; text)	$+0.064$ $[0.053, 0.073]$
P3$\times$P5	-0.175 $[-0.236, -0.112]$	-0.070 $[-0.080, -0.059]$
P2$\times$P4	-0.058 $[-0.117, 0.004]$	$+0.000$ $[-0.010, 0.010]$
P1$\times$P2	$+0.008$ $[-0.052, 0.069]$	$+0.005$ $[-0.005, 0.015]$

is carried by the other axes of the vector: zero agent-side messages, graceful degradation under coordinator denial (below), and no second infrastructure to keep alive. The sandbox's medium bill is real and must not be hidden: the field arms pay $\sim 10^5$ cell-deliveries per unit time against gossip's 3.8×10^3 messages — the price of a physics nobody has to route.

H1, sandbox face (Fig. B2a). Sweeping demand mass D from 2 to 12 baseline units (injury): the hormonal arm's median resolve time grows near-linearly (26 $\to$ 148, slope $\approx$ 12 per unit D) with production scaling $u_{\rm peak}/u_0 = 3.5$–3.8; the fixed arm cannot scale ($u_{\rm peak}/u_0 = 1.0$ throughout), fails 33% at $D = 4$ and 100% from $D = 8$. The tracking arm's own failures at $D \geq 10$ (17%, 40%) are honest: they are serving-and-migration limits of the grid, not loop limits — production was still scaling when the deadline expired.

H2, fibrosis (Fig. B2b, bars). Chronic demand clearing at $t = 6\tau$: with P5 the pool stands down to 1.03$\times$ baseline and production to 1.00$\times$; without P5 the pool ratchets to 3.07$\times$ and production to 3.10$\times$ and stays there — Fig. 3(b)'s fibrotic shift, reproduced with spatial routing in the loop, in a measurable fraction of runs (failure 13% versus 3%).

H2b, adversarial recruitment (Fig. B2b, lines). Sweeping false-signal injection rate $\nu = 0 \to 0.2$: with the P4 co-signal gate, false burn is *zero* at every ν; without it, burn rises from 12 (at $\nu = 0$ — friendly fire at stale halos) to 78 pool mass at $\nu = 0.2$. Misallocated holding is similar in both arms ($\sim$5–

Table B4. The six Phase II arms at matched information and budget (30 seeds; time in production delays τ; deadline 8τ). Resolve = median time to clear demand (censored at run end). Cost components per unit simulated time on the 24×24 grid (infection scenario; an emission bills $G^2 = 576$ cell-deliveries; all arms additionally hold $3G^2$ words of agent state).

	failure / resolve (τ)			cost per unit time			
arm	injury	infection	chronic	emissions	deliveries (10^3)	messages	medium state (words)
hormonal (P1–P5)	0.00 / 3.6	0.10 / 4.8	0.03 / 4.8	191	110	0	2304
fixed (no P3)	0.73 / >8	1.00 / >8	1.00 / >8	191	110	0	2304
measured-error	0.00 / 3.5	0.10 / 4.7	0.03 / 4.7	191	110	1.5	2304
back-pressure	0.00 / 0.9	0.00 / 0.9	0.00 / 0.9	1.5	0.9	0	—
gossip	0.00 / 0.9	0.00 / 0.9	0.00 / 0.9	—	—	3848	—
oracle	0.00 / 0.3	0.00 / 0.3	0.00 / 0.3	—	—	0	— †

†the oracle instead reads global state (G^2 words per read, continuously): a coordinator hidden in the instrumentation, by design the upper bound.

11): P4 gates *expenditure*, not attention — the guard's FPR claim holds, and its cost shows up as the throughput tax of Table B3.

H3, coordinator denial and churn. The oracle arm under two outages of its coordinator channel ((0, 30) and (50, 90), 35% of the run): unmet demand rises 6.5× over the clean oracle (0.104 versus 0.016) — to *worse than the hormonal arm* (0.077), which has no coordinator to deny. The hormonal arm under churn (random removal of 10–50% of off-site responders every τ) does not degrade: failure 0.10 → 0.03, unmet 0.077 → 0.073 — turnover prunes the wandering pool, keeps the residual honest, and sustained delivery continues; the loop is robust to, even mildly aided by, the constant turnover its biological reference runs natively.

B3 What the pilot changes in the programme

(i) *The settling budget is the binding constraint, and the gate volume is a sliver (11.5%), not a majority.* The termination criterion stands (a realistic-delay region passes with majority robustness), but the definitive Phase I must either justify denominating settling in $\max(\tau, \beta^{-1}, m^{-1})$ — the slowest loop timescale, which the biology arguably supplies — or preregister the honest volume. Sobol says the envelope is set by timescales and demand, not loop gain: y is nearly free to leave unidentified.

(ii) *Residual sensing latency, not noise, spends the budget* (Table B1): a $0.25\,\tau$ measurement delay removes most of the gate volume while $\sigma = 0.1$ jitter costs nothing. The Phase IV substrate requirement "medium clears on the signal timescale" is quantitative, not decorative.

(iii) *The primitives buy robustness and terminability, not throughput.* P3-only beats the full stack on failure in the benign scenario; the guard and antagonist each cost a few points there and pay back exactly where H2 (fibrosis: 3.07× versus 1.03×) and H2b (false burn 78 versus 0) bill them. The preregistered outcome vector must weight termination and adversarial axes, as Section 4.2 already commits.

(iv) *P3's equivalence, not superiority, at pilot scale*: the measured-error integral controller on site reports matches the physical residual within 2% at 1.5 messages per unit time. The hormonal arm's advantage appears on the denial and infrastructure axes (H3: oracle-under-outage 0.104 unmet versus hormonal 0.077), and the definitive comparison must report all five cost components to make that visible.

(v) *H3's graceful-degradation claim survives contact*: no coordinator to deny, robust to 50% churn. The churn arm even improves slightly — turnover keeps the residual honest — which the definitive runs should confirm at scale rather than lean on.

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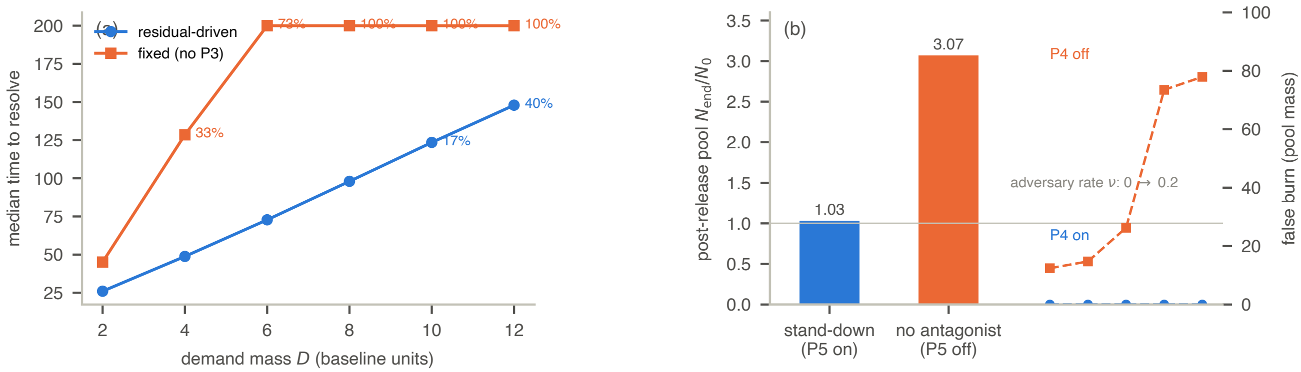

Figure B2. Phase II pilot. (a) H1: median time to resolve versus demand mass D (injury scenario; deadline 8τ): the residual- driven arm scales production ($u_{\mathrm{peak}}/u_0 = 3.5$–$3.8$) and resolves near-linearly, with graceful degradation at $D \geq 10$; fixed production cannot scale and fails outright (percentages: failure rate). (b) H2/H2b: bars — post-release pool N_{end}/N_0 in the chronic scenario with the antagonist (P5) on and off, the fibrotic shift; lines — false burn (expended pool mass) versus adversary rate ν with the P4 co-signal gate on and off.

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